TEST DOCUMENTATION

PROJECT: Traduce Señas

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Introduction

The test documentation brings together the strategy, the test plan, the test cases, and the results obtained for the Traduce Señas software. Its purpose is to demonstrate with evidence that the system meets the requirements of the specification report and to identify the aspects that have not yet been verified.

The structure of this document takes as reference the ISO/IEC/IEEE 29119-3 standard on test documentation (International Organization for Standardization [ISO], 2013) and the ISO/IEC 25010 quality model (ISO, 2011).

Test objectives

- Verify that each microservice meets its business logic and security rules.
- Verify that the services integrate correctly through the gateway and events.
- Validate the critical functional requirements (authentication, translation, history, alphabet, statistics).
- Evaluate the non-functional requirements for accessibility, performance, and security.

Test strategy

The test pyramid is applied: many fast unit tests at the base, fewer integration and contract tests in the middle, and few end-to-end tests at the top.

Table 1
Test levels

Level	Objective	Tool	Target
Unit	Domain logic and use cases in isolation, with fake repositories	node:test (backend); Jest + jest-expo (frontend)	≥ 80% of lines; ≥ 90% in domain/
Integration	Adapters with real PostgreSQL, MongoDB, and RabbitMQ	node:test + Docker Compose	Happy path and 2–3 errors per adapter
Contract	HTTP responses comply with the OpenAPI contract	YAML contracts from 07-api	100% of public routes
End-to-end	Complete user flows through the gateway	Playwright in CI against the backend in Docker Compose; HTTP client / manual test	Critical flows
Acceptance	Validation with real users (deaf and hearing people)	Guided pilot test	Use cases CU-01 to CU-06

Level	Objective	Tool	Target
Accessibility	Contrast, target size, and screen readers	Accessibility Scanner, TalkBack, manual review	WCAG 2.1 AA
Performance	Latency and performance of critical services	k6	P95 < 300 ms with 50 requests/s

Test plan

Items to test

- api-gateway and the five microservices (backend version 1.4.0, branch main, commit a962982).
- Shared package @traduce/shared.
- Client application on Android and in a desktop browser, with unit tests for its logic: validations, API client, authentication flows, and recognition (frontend version 1.3.1, branch main, commit a5e231d).

Items out of scope for this iteration

- Load tests on the production server (traducesenas.tech), so as not to affect users.
- Publication in application stores.
- Recognition accuracy of the 20 words in the LSC-54 set (pending the measurement described in the recognition section).

Test environment

Table 2
Environment used

Element	Detail
Equipment	Computer with Windows 11, Node.js 24, and Docker Desktop
Backend	Docker Compose: gateway, 5 services, 5 PostgreSQL 16, MongoDB 7, RabbitMQ 3
Client	Expo web in Chrome; Expo Go on Android
Data	Users and sign test data created by the tests themselves; no real data

Acceptance and suspension criteria

- **Acceptance:** 100% of the automated tests passing and all critical cases passed.
- **Suspension:** execution stops if the build fails or if a critical security test fails.
- **Resumption:** after correcting the defect and re-running the full suite.

Roles

Table 3
Test responsibilities

Role	Responsible	Tasks
Test lead	Juan Mauricio Suaza Solórzano	Planning, backend tests, integration, and reporting
UI tester	Luis Alejandro Duarte Aldana	Screen and accessibility tests
Functional tester	Juan Camilo Fierro Esquivel	Acceptance cases and pilot test

Results of the automated tests

The automated suites were run with “npm test” in each repository. In the backend, the command builds the shared package and runs the tests of all services with the native Node.js test runner; in the frontend, it runs Jest with the Expo preset (jest-expo). The current result, from the run of November 4, 2026, is: in the backend, 918 tests passed, 0 failed and 13 skipped because they require PostgreSQL on the local machine; in the frontend, 469 tests in 46 suites, all passed. The first frontend run had a 5 s timeout in AchievementUnlockAlerts.test.tsx with the machine under load; it passed after raising the wait time to 30 s (“--testTimeout=30000”). Table 4 keeps the October 16, 2026 run as a historical reference:

Table 4
Summary of the automated run of October 16, 2026

Indicator	Backend	Frontend
Tests executed	683	79
Suites (groups)	141	8
Passed	681	79
Failed	0	0
Skipped / cancelled	2 / 0	0 / 0
Total duration	22 seconds	3 seconds

Note. Run of October 16, 2026. Backend: branch develop, commit 2993301; the 2 skipped tests need a test PostgreSQL database (variable PROFILE_TEST_DATABASE_URL). Frontend: branch chore/frontend-unit-tests, commit 6219c0d.

Table 5
Test coverage by component

Component	Test files	Aspects verified	Line coverage
@traduce/shared	7	JWT provider, authMiddleware, requireRole, requirePermission, error handler, configuration, event validators, exponential retry, in-memory bus, and RabbitMQ	99.5%

Component	Test files	Aspects verified	Line coverage
api-gateway	3	Prefix-based routing, CORS with allowlist, configuration, public routes, rate limiting	100%
iam-service	33	Registration with email verification, login according to status and with Google or Facebook, password recovery, JWT with roles and permissions, account change and deletion (also with an email code), event delivery through an outbox, email templates	89.3%
recognition-service	17	Translations, history, sign templates (ADMIN only), samples, consumption of lexicon events with contract, and rejection of writes from deleted accounts	86.0%
lexicon-service	5	Catalog, alphabet, search, categories, publication, and withdrawal of signs	83.5%
analytics-service	6	Event logging, reports, event handlers, and atomic erasure of an account's data	95.1%
profile-service	27	Profile, preferences with permissions and creation of a missing profile, notifications, achievements under concurrent updates, account deletion, and profile recovery	83.5%
Frontend (client application)	46 Jest (plus 3 Playwright E2E)	Email and password validation, API client (errors, token sent only to the API, single expired-session notice), confirm-email and password-recovery flows, resend countdown, hand normalization, and KNN classifier	4.5%

Note. Line coverage figures were measured on October 16, 2026 with the built-in node:test coverage in the backend (88.5% in total) and with “jest --coverage” in the frontend. That run predates the tests described below and has not been repeated. The test-file column of Table 5 is from November 4, 2026 and counts the files of each component on the main branch.

Tests added after that run. In the frontend (current version 1.3.1), Jest tests were added for the services, the authentication forms, the vision classifiers and the template stores, plus an end-to-end suite with Playwright (e2e/ folder: auth, alphabet and history). The suite runs with “npm run e2e” on the exported web app, against the backend started with Docker Compose, with a single worker (the gateway limits logins per minute), a phone-sized viewport and the browser's fake camera; in CI it runs against the backend in compose. In the backend (profile-service), tests were added for daily challenges (domain, use cases, API and real PostgreSQL), challenge-point medals and the emails sent to administrators.

Functional test cases

The following cases verify the critical requirements. The “Passed” status indicates that

the case was executed successfully; “Pending” indicates that it has not yet been executed in the corresponding environment.

Table 6
Functional test cases

ID	Requirement	Case	Input data	Expected result	Status
CP-01	FR-001, FR-003	Registration with email verification	New email, valid data, terms accepted	Account INACTIVE; code sent; activated after verification	Passed (automated and E2E)
CP-02	FR-002	Registration without accepting terms	terms_accepted = false	400; account not created	Pending (no specific automated test)
CP-03	FR-005	Valid login	Correct credentials, account ACTIVE	200 with JWT token containing roles	Passed (automated and E2E)
CP-04	FR-005	Login with INACTIVE or BLOCKED account	Unverified or blocked account	Access denied with a clear message	Passed (automated)
CP-05	FR-004, FR-010	Recovery with expired or used code	Code expired / already used	Rejection; attempts are counted	Passed (automated)
CP-06	RNF4	Rate limit on /api/auth	Burst of requests from one IP	429 Too Many Requests	Passed (automated)
CP-07	FR-103	User without ADMIN role imports templates	Token with USER role	403 Forbidden	Passed (automated)
CP-08	FR-401, FR-406	Translation saved to history	Sign recognized, user authenticated	201; appears in history	Passed (E2E with Docker)
CP-09	FR-802	Achievement “First translation”	First translation by a user	Achievement unlocked by the TranslationProduced event	Passed (E2E with Docker)
CP-10	FR-701	Usage event recorded	Translation saved	Event in analytics-service	Passed (E2E with Docker)
CP-11	FR-405	Sign not recognized	Gesture outside the vocabulary	Visual notice; no word is shown	Pending (manual test on device)

ID	Requirement	Case	Input data	Expected result	Status
CP-12	FR-304, FR-310	Alphabet query	GET /api/lexicon/alphabet	Only LSC letters in ACTIVE status	Passed (automated)
CP-13	FR-406	Private history	User A queries the history of B	Sees only their own translations	Pending (manual test)
CP-14	FR-806, FR-807	Daily challenge completed	User who saves as many translations as the day's target	Progress up to the target; reward stored only once; CHALLENGE_COMPLETED notice	Passed (automated, run of November 4, 2026)
CP-15	FR-808	Challenge-point medal	User who accumulates challenge points	GET /api/profile/challenges/rewards/summary returns the total; the medal is granted once	Passed (automated, run of November 4, 2026)
CP-16	FR-409, FR-707	History by sessions and own URL	Open the history by its URL with several saved translations	One card per session; the screen reloads without losing the route	Pending (the history E2E covers saving and deleting a translation, but not the sessions or the own URL)
CP-17	FR-410	30-day retention	Translation older than 30 days	The translation no longer appears in the history	Pending (manual test)
CP-18	FR-014	Delete account	Authenticated user	Data deleted in one transaction and usage events anonymized	Passed (automated)
CP-19	FR-602	Sample without consent	Sample without consent field	400; the sample is not saved	Passed (automated)
CP-20	FR-706	Statistics with USER role	Token without permission	403 Forbidden	Passed (automated)

Note. The end-to-end test with Docker (registration → login → translation → achievement and usage event) was executed on October 5, 2026.

Non-functional tests

Table 7
Status of the non-functional tests

Attribute	Test	Criterion	Status
Accessibility	Color contrast of the design system	≥ 4,5:1	Passed: tokens adjusted in the design system

Attribute	Test	Criterion	Status
Accessibility	TalkBack navigation in main flows	Complete flows without blocking	Pending
Security	Mandatory JWT secret of 32+ characters	The service does not start without it	Passed (automated)
Security	Databases and broker not exposed to the network	Ports only on 127.0.0.1	Passed (docker-compose review)
Performance	P95 latency of login, translations, and lexicon	< 300 ms with 50 requests/s	Pending (k6)
Performance	Frames per second in capture	≥ 15 FPS on mid-range devices	Pending
Fault tolerance	RabbitMQ outage and recovery	The bus reconnects and retries	Passed (automated)

Sign recognition tests

Recognition accuracy is not measured with unit tests but with an evaluation protocol: samples are taken from people who did not take part in building the templates, the same method used by the application is applied (normalization and Dynamic Time Warping), and for each sign the percentage of correct hits, the percentage of misrecognized words, and the signs with which it is confused are calculated.

For the 20 new words, the public LSC-54 set is used (Mora-Zarate et al., 2025), separating training and test signers. The acceptance target is an accuracy of at least 85% before publishing the templates. This measurement is pending because the data download was stopped due to restrictions on the source server.

Defects found

Table 8
Defect log

ID	Description	Severity	Status
DEF-01	The LSC-54 set downloader could skip entire videos without warning	High	Fixed
DEF-02	Achievement counter not atomic: increments were lost with simultaneous translations	Medium	Fixed: atomic update with ON CONFLICT DO UPDATE
DEF-03	Unescaped username in the HTML of the recovery email	High (security)	Fixed: escapeHtml function

ID	Description	Severity	Status
DEF-04	GET /api/profile/me failed if the profile did not yet exist	Medium	Mitigated: a missing profile is created when preferences are saved and there is a recovery procedure; the query still returns 404 while it is missing
DEF-05	lexicon event contract test failed (305 of 306)	Low	Fixed: the test passes in the current suite
DEF-06	The letter L is never recognized: its hand shape matches the D or G rule first	Medium	Open (found by the frontend unit tests)
DEF-07	Two iam-service migrations with the same number (010) when integrating account deletion	Medium	Fixed: renumbered to 012 before deployment
DEF-08	A recognition-service test returned 500 because of the new deleted-account guard	Low	Fixed: the test uses a simulated connection

Note. DEF-01 to DEF-05 come from the code and security audit of phase 4 of the microservices migration; DEF-06 to DEF-08 come from the frontend unit tests and the code review of October 16, 2026.

Conclusions

The backend passes its automated suite: 918 tests passed and none failed; 13 are skipped on the local machine because they need a PostgreSQL database. The frontend passes its 469 Jest tests (46 suites), and its 3 end-to-end Playwright tests run in CI. Line coverage was last measured on October 16, 2026: 88.5% in the backend, above the 80% target, and 4.5% in the frontend, which is still low. Jest has no coverage threshold configured, although ADR-006 sets an 80% gate. Continuous integration already runs in both repositories. The critical end-to-end flows work with all services deployed in Docker. Pending items are extending the frontend tests to the screens, the k6 performance tests, the screen reader tests, the measurement of recognition accuracy for the new words, and the pilot test with real users, which are scheduled for the next iteration.

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